Addition and subtraction of algebraic fractions

1 Simplify:

a $\frac{2x}{7}+\frac{7x}{7}$ **b** $\frac{3x}{4}+\frac{2x}{4}$ **c** $\frac{2x}{3}+\frac{10x}{3}$ **d** $\frac{3x}{12}+\frac{12x}{12}$

e $\frac{-8x}{9}+\frac{4x}{9}$ **f** $\frac{-3x}{14}+\frac{-2x}{14}$ **g** $\frac{8x}{2}-\frac{3x}{2}$ **h** $\frac{13x}{3}-\frac{8x}{3}$

i $\frac{7x}{5}-\frac{10x}{5}$ **j** $\frac{31x}{6}-\frac{7x}{6}$ **k** $\frac{25x}{24}-\frac{5x}{24}$ **l** $-\frac{4x}{7}-\frac{5x}{7}$

2 Simplify:

a $\frac{2x}{3}+\frac{x+3}{3}$ **b** $\frac{7x}{8}+\frac{5x+2}{8}$

c $\frac{-2x}{7}+\frac{4x-3}{7}$ **d** $\frac{5x-1}{3}-\frac{2x+3}{3}$

e $\frac{4x-10}{3}+\frac{-5x+10}{3}$ **f** $\frac{-6x-8}{5}+\frac{-3x-5}{5}$

g $\frac{4x^2+11}{6}-\frac{3x^2-4x}{6}$ **h** $\frac{2x^2-3}{5}+\frac{3x^2+6}{5}$

i $\frac{-5x^2-2x}{3}+\frac{-3x^2-2x}{3}$ **j** $\frac{2x-7x^3}{4}-\frac{-x^2-3x^3+1}{4}$

3 Find the lowest common denominator for the following pairs of algebraic fractions:

a $\frac{w}{2}$ and $\frac{w}{6}$ **b** $\frac{s}{3}$ and $\frac{s}{6}$ **c** $\frac{x}{2}$ and $\frac{y}{3}$ **d** $\frac{r}{5}$ and $\frac{s}{3}$

4 Consider the algebraic fractions $\frac{2m}{5}$ and $\frac{3m}{6}$.

a Find the lowest common denominator. **b** Hence, simplify $\frac{2m}{5}+\frac{3m}{6}$.

5 Complete the following:

$$\frac{3x}{5}+\frac{2x}{\square}=\frac{\square}{35}$$

6 Simplify:

a $\frac{2x}{5}+\frac{4x}{7}$ **b** $\frac{2x}{14}+\frac{14x}{21}$ **c** $\frac{4x}{5}-\frac{x}{7}$ **d** $\frac{4x}{45}-\frac{x}{18}$

e $-9x-\frac{5x}{7}$ **f** $\frac{3x}{5}-\frac{2x}{7}$ **g** $-3x-\frac{3x}{8}$ **h** $\frac{11x}{14}+\frac{7x}{21}$

i $\frac{x}{15}+\frac{3x}{35}$ **j** $\frac{4x^2}{3}-\frac{5x^2}{2}$

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7 Simplify:

a $\frac{2x}{3} + \frac{2x+5}{9}$

b $\frac{8x}{13} + \frac{3x+4}{39}$

c $\frac{3x}{10} - \frac{4x+3}{16}$

d $\frac{4x}{9} + \frac{2x-3}{36}$

e $\frac{8x}{11} - \frac{2x-3}{44}$

f $\frac{y}{10} - \frac{5y+1}{25}$

g $\frac{7x}{8} - \frac{5x+6}{20}$

h $\frac{x+3}{3} + \frac{3x-2}{12}$

i $\frac{x+3}{3} - \frac{5x-7}{12}$

j $\frac{2x+3}{4} - \frac{2x+2}{12}$

k $\frac{5x+8}{6} + \frac{x+2}{2}$

l $\frac{3x+4}{6} + \frac{3x+4}{5}$

m $\frac{2x+2}{4} - \frac{x+3}{8}$

n $\frac{6x+7}{3} - \frac{3x-5}{9}$

o $\frac{6x^2}{7} + \frac{x^2+5x}{21}$

p $\frac{2x^2}{7} + \frac{3x^2+2x}{28}$

8 Simplify:

a $\frac{3}{x} + \frac{5}{y}$

b $\frac{c}{ab} + \frac{2}{bc^2}$

c $\frac{2x^2+4y}{y} - \frac{x-3}{x}$

d $\frac{4}{x} + \frac{2}{x-2}$

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